

Variable and model selection

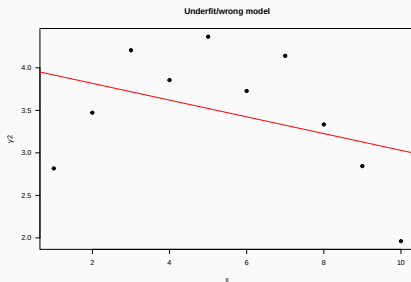
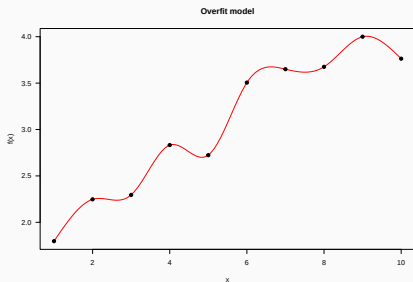
Francisco Rodríguez-Sánchez

<https://frodriguezsanchez.net>

Overfitting and balanced model complexity

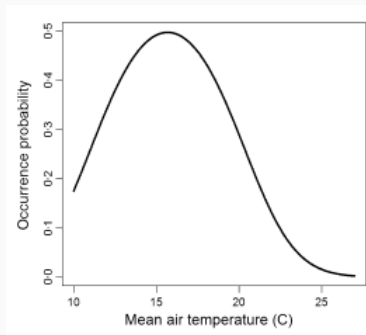
On one hand, we want to **maximise fit**.

On the other hand, we want to **avoid overfitting** and overly complex models.

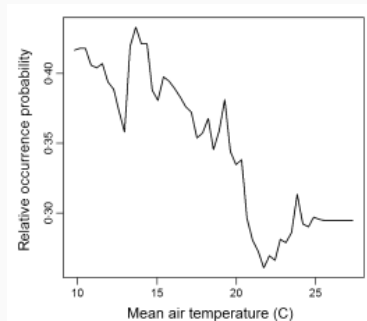


Overfitting and balanced model complexity

GLMM



Random forests



Wenger & Olden (2012)

Overfitted models will work badly on new data



- **Cross-validation** (k-fold, leave one out...)

Evaluating models' predictive accuracy

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- **Information Criteria:**
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 - WAIC...
- All these methods have flaws!

AIC (Akaike Information Criteria)

$$AIC = -2 * \text{LogLikelihood} + 2K$$

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- AIC **biased** towards complex models.
- **AICc** recommended with 'small' sample sizes ($n/p < 40$). But see [Richards 2005](#)

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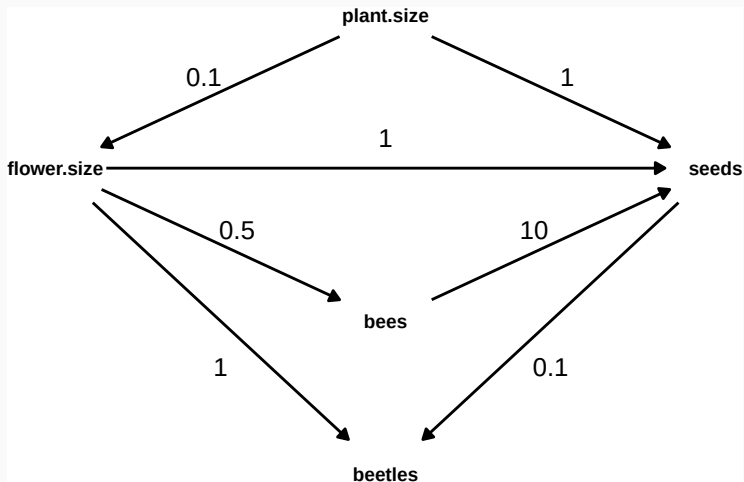
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- Predictive criteria don't help for **causal inference**.

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Predictive criteria don't help to choose correct causal model

Making good predictions doesn't require accurate causal model

Model	AIC	R2
m.flower	933.3	0.5
m.flower.plant	913.2	0.6
m.flower.plant.bees	899.1	0.7
m.flower.plant.bees.beetles	829.9	0.8

'Best model' (based on AIC or R2) not good for causal inference

So which variables should enter
my model?

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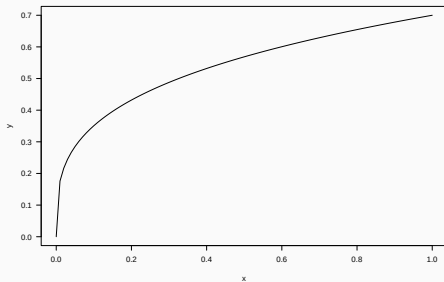
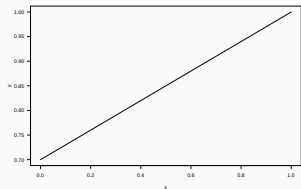
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- For predictors with large effects, **consider interactions**.

Think about the shape of relationships

$$y \sim x + z$$

Really? Not everything has to be linear! Actually, it often is not.

Think about shape of relationship.



Removing predictors

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- Should we remove highly collinear predictors?

Collinearity isn't a disease that needs curing

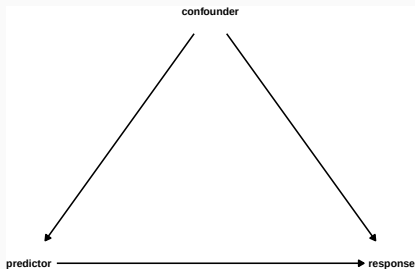
The Meaning of Multiple Regression and the Non-Problem of Collinearity

The VIF Score. What is it Good For? Absolutely Nothing

Additional caution regarding rules of thumb for VIF

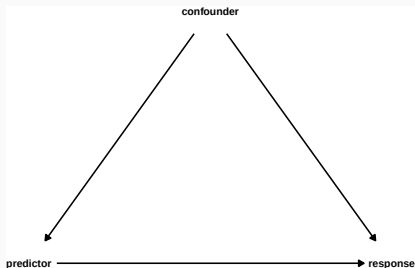
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Removing predictors

- If **predictive** goal, just consider predictive performance
- If **explanatory** goal, consider causal structure of your problem (DAG)



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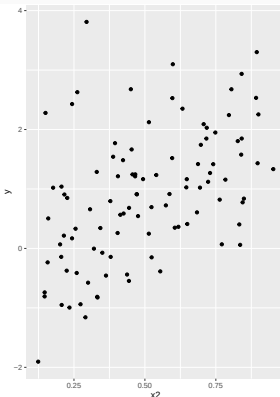
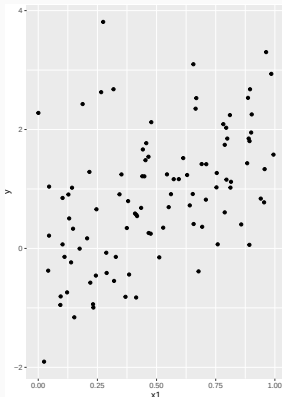
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- Mundry & Nunn (2009) Stepwise Model Fitting and Statistical Inference: **Turning Noise into Signal Pollution**. Am Nat.
- This includes **stepAIC** (e.g. Dahlgren 2010; Burnham et al 2011; Hegyi & Garamszegi 2011).

Simpler (best) model provides biased causal estimates

Simulate response depending on two correlated variables (Hartig 2022)

```
x1 = runif(100)
x2 = 0.8*x1 + 0.2*runif(100)
y = x1 + x2 + rnorm(100)
```



Simpler (best) model provides biased causal estimates

Simulate response depending on two correlated variables (Hartig 2022)

```
fullmodel = lm(y ~ x1 + x2)
```

Call:

```
lm(formula = y ~ x1 + x2)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.8994	-0.6821	-0.1086	0.5749	3.3663

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.1408	0.2862	-0.492	0.624
x1	1.2158	1.5037	0.809	0.421
x2	0.8518	1.8674	0.456	0.649

Residual standard error: 0.9765 on 97 degrees of freedom

Multiple R-squared: 0.237, Adjusted R-squared: 0.2212

F-statistic: 15.06 on 2 and 97 DF, p-value: 2.009e-06

Simpler (best) model provides biased causal estimates

Check for Multicollinearity

High Correlation

Term	VIF	VIF 95% CI	adj. VIF	Tolerance	Tolerance 95% CI
x1	19.07	[13.34, 27.45]	4.37	0.05	[0.04, 0.07]
x2	19.07	[13.34, 27.45]	4.37	0.05	[0.04, 0.07]

Simpler (best) model provides biased causal estimates

```
simplemodel = MASS::stepAIC(fullmodel, trace = 0)
```

Call:

```
lm(formula = y ~ x1)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.9047	-0.6292	-0.1019	0.6077	3.3394

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.04633	0.19670	-0.236	0.814
x1	1.88350	0.34295	5.492	3.13e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9725 on 98 degrees of freedom

Multiple R-squared: 0.2353, Adjusted R-squared: 0.2275

F-statistic: 30.16 on 1 and 98 DF, p-value: 3.134e-07

Automated model selection (dredge)

Simulating data with 10 random predictors

```
dat <- data.frame(y = rnorm(100),  
                  x = matrix(runif(1000), ncol = 10))
```

y	x.1	x.2	x.3	x.4	x.5	x.6	x.7	x.8	x.9	x.10
-0.1	0.6	0.6	0.3	0.8	0.2	0.0	0.4	0.4	0.3	0.2
0.8	0.4	0.4	0.9	0.2	0.5	0.1	0.6	0.2	0.0	0.0
-0.5	0.0	0.3	0.4	0.3	0.1	0.1	0.9	0.9	0.5	0.8
-0.6	0.7	0.7	0.4	0.5	0.2	0.7	0.7	0.8	0.5	0.3
0.7	0.0	0.6	0.9	0.1	0.2	0.4	0.8	0.6	0.6	0.1
-0.1	0.4	0.2	0.9	0.4	0.6	0.5	0.9	0.1	0.8	0.8

Automated model selection

Running `MuMIn::dredge` with 10 random predictors

```
full.model <- lm(y ~ ., data = dat)
dd <- MuMIn::dredge(full.model)
```

Best model:

Parameter	Coefficient	SE	p
(Intercept)	-1.50	0.36	0.00
x.2	0.78	0.36	0.03
x.5	0.59	0.32	0.07
x.6	0.61	0.35	0.09
x.9	0.87	0.34	0.01

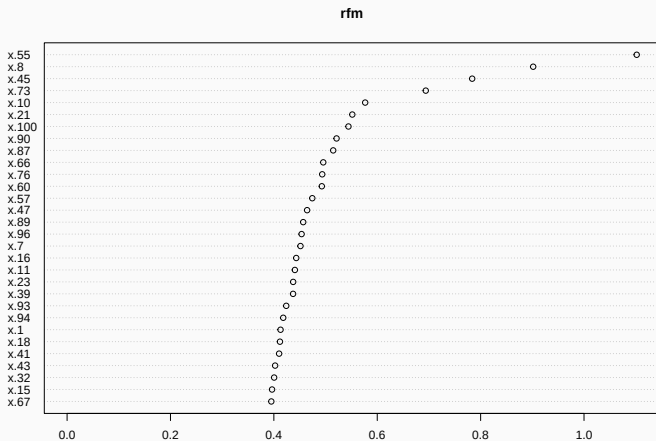
“Let the computer find out” is a poor strategy and usually reflects the fact that the researcher did not bother to think clearly about the problem of interest and its scientific setting

Burnham and Anderson 2002

Variable importance in machine learning

Random forest on 100 random predictors

```
dat <- data.frame(x = matrix(runif(50000), ncol = 100), y = runif(500))  
rfm <- randomForest::randomForest(y ~ ., data = dat)  
varImpPlot(rfm)
```



Other common bad practices

- Testing bivariate relationships before building multivariable model

Heinze & Dunkler 2016

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- Testing bivariate relationships before building multivariable model
- Removing non-significant predictors

Heinze & Dunkler 2016

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- Always **keep 'core' predictors** (based on previous knowledge)

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- Always **keep 'core' predictors** (based on previous knowledge)
- If ratio sample size/number of predictors is low (<10 EPP), avoid variable selection (too unstable)
- If performing variable selection, always **assess stability** (bootstrap, etc)

Heinze et al 2018

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3. If > 1 model have similar support, consider model averaging (or blending).
4. Always check fitted models thoroughly
5. Always report effect sizes